

Case Study - Group 4

Research question

The research question is: **"Is Gender a Significant Factor in the Increase of the Number of Heterosexual Partners in the Last Decade?"**

1. Try to explain why this research questions might be of interest and to whom.

Why this research question is interesting

This research question matters because the number of heterosexual partners is often used as an indicator of broader changes in sexual behaviour, relationship patterns, and sexual health risk. If the number of partners has increased over time, that may suggest shifts in social norms around sex, dating, and intimacy. Looking specifically at gender makes the question even more relevant, because men and women have historically been judged differently for similar sexual behaviour, and those differences may be changing.

Why gender is an important factor

Gender is central here because sexual behaviour is shaped not only by personal choice, but also by social expectations and cultural norms. Traditionally, men have often been given more social acceptance for reporting multiple sexual partners, while women have faced greater stigma for the same behaviour. If the gap between men and women is narrowing, that could indicate changing attitudes toward gender roles and sexuality. If the gap remains large, that may suggest that older double standards still influence behaviour or at least how behaviour is reported.

So this question is not just about counting partners. It is also about understanding whether changes over time are happening equally for men and women, or whether one gender is driving the trend more than the other.

Why this matters academically

For researchers in sociology, gender studies, psychology, and public health, this question is useful because it connects private behaviour to wider social change. It can help explore:

- whether sexual lifestyles are becoming more similar across genders
- whether traditional gender norms are weakening
- how sexual attitudes translate into actual behaviour
- whether reported increases reflect real behavioural change or changing willingness to disclose sensitive information

Why this matters for public health and policy

This topic is also important for public health professionals and policymakers. The number of sexual partners is relevant to discussions about:

- sexually transmitted infections
- HIV risk

- access to sexual health services
- sex education needs
- targeted health campaigns

If patterns differ by gender, then interventions may need to be designed differently for men and women. For example, if one group has experienced a sharper increase in partner numbers, that could influence where prevention resources or educational messaging should be focused.

Who would be interested in this research

This research could be of interest to several groups.

- Academics would be interested because it helps explain changes in sexual behaviour and gender relations over time.
- Public health officials would care because partner numbers are linked to sexual health planning and risk monitoring.
- Policymakers may find it useful when designing sex education programs or allocating funding for sexual health services.
- Healthcare providers could use this knowledge to better understand behavioural trends in the population they serve.
- It may also interest the general public, especially in discussions about whether society is becoming more sexually liberal and whether expectations for men and women are becoming more equal.

2. Locate relevant data file and accompanying documentation in the “stata” folder.

The relevant data files are the Stata `.dta` files.

- For Natsal-2, the data file is: `CaseStudyNSSAL\UKDA-5223-stata\NSSAL_2000.dta`
- For Natsal-3, the data file is: `CaseStudyNSSAL\UKDA-7799-stata\NSSAL_2010.dta`

3. Perform any necessary transformations of data format type.

```
# 1. Clean column names
# 2. Keep only important variables for NSSAL 2000
# 3. Keep only important variables for NSSAL 2010
# 4. Rename variables so both datasets match
# 5. Add survey wave
# 6. Combine both datasets
# 7. Keep an even smaller final analysis dataset
# 8. Recode missing values
```

```
# 9. Restrict to comparable age range
# 10. Create readable labels
# 11. fix gender capitalization
```

4. Identify variables of interest and specify their type.

Purpose	Column(s)	Why important
Respondent ID	<code>Sserial / sin2</code> — identifier	Identifies each participant
Gender/sex	<code>rsex</code> — nominal / binary	Main independent variable: male/female
Age	<code>dage</code> — scale; <code>agrp</code> , <code>agrp2</code> — ordinal	Needed because Natsal-2 and Natsal-3 must be compared in the same age range
Survey weight	<code>final_wt</code> , <code>total_wt</code> — scale / weight variables	Needed if we want representative results
Survey design	<code>psu/psu_scrm</code> , <code>strata</code> , <code>totalpsu</code> , <code>totalstr</code> — design variables / nominal	Important for proper survey-based statistics
Heterosexual partners, lifetime	<code>hetlife</code> — scale/count; <code>hettot</code> — ordinal/grouped	Likely the main outcome variable
Heterosexual partners, last 5 years	<code>het5yrs</code> — scale/count; <code>het5yrg</code> — ordinal/grouped	Useful if the question is about recent change
Heterosexual partners, last year	<code>het1yr</code> — scale/count; <code>het1yrg</code> — ordinal/grouped	Good alternative outcome
Heterosexual partners, last 3 months	<code>het3mnt</code> — scale/count; <code>het3mntg</code> — ordinal/grouped	Less central, but possible
New heterosexual partners	<code>hetnewp</code> — nominal/binary; <code>hetnew</code> — scale/count; <code>hetnew2</code> — nominal/binary; <code>hetnew3</code> — ordinal/grouped	Useful if the question is about increase/new partners
Education	<code>educ</code> , <code>educ2</code> , <code>educ3</code> , <code>exams2</code> — ordinal	Control variable

Marital/relationship status	marstat, ms, ms2, ms3 — nominal	Very important control
Region	gor, gor2, stdreg, london — nominal	Control variable
Sexual orientation / eligibility	het, hom/sam, scelig/scelign2 , sexcomp/sexcomp2 — nominal/binary	Helps decide who should be included/excluded

5. In case a variable is ordinal or nominal, list the possible categories and their meaning.

Nominal variable

Nominal means the values are just **categories with no natural order**.

Pos. = 10 **Variable = rsex** **Variable label = respondent's sex**

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for rsex

Value = 1.0 Label = male
Value = 2.0 Label = female

Pos. = 1104 **Variable = totalstr** **Variable label = strata for the core sample and ethnic minority boost sample combined**

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for totalstr

Value = 1.0 Label = inner london
Value = 2.0 Label = outer london
Value = 3.0 Label = rest of britain
Value = 4.0 Label = ethnic boost, stratum 1
Value = 5.0 Label = ethnic boost, stratum 2

Pos. = 389 **Variable = marstat** **Variable label = marital status at present**

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for marstat

Value = 1.0 Label = married & living with spouse
Value = 2.0 Label = living with a partner of opposite sex as a couple
Value = 3.0 Label = living with a partner of same sex as a couple
Value = 4.0 Label = widowed
Value = 5.0 Label = divorced
Value = 6.0 Label = seperated
Value = 7.0 Label = single & never been married
Value = 9.0 Label = not answered

Pos. = 798 **Variable = ms** **Variable label = marital status, grouped**

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for ms

Value = 1.0 Label = married
Value = 2.0 Label = cohabitation, opposite sex
Value = 3.0 Label = cohabitation, same sex

Value = 4.0	Label = widowed
Value = 5.0	Label = divorced/separated
Value = 6.0	Label = single
Value = 9.0	Label = not answered

Pos. = 799 Variable = **ms2** Variable label = **marital status, grouped**

This variable is *numeric*, the SPSS measurement level is **NOMINAL**

Value label information for ms2

Value = 1.0	Label = married & living with spouse
Value = 2.0	Label = cohabitation (opposite sex & same sex)
Value = 3.0	Label = previously married (widowed, divorced, separated)
Value = 4.0	Label = single and never been married
Value = 9.0	Label = not answered

Pos. = 800 Variable = **ms3** Variable label = **marital status, grouped**

This variable is *numeric*, the SPSS measurement level is **NOMINAL**

Value label information for ms3

Value = 1.0	Label = married & living with spouse
Value = 2.0	Label = cohabitation, opposite sex
Value = 3.0	Label = cohabitation, same sex
Value = 4.0	Label = previously married (widowed, divorced, separated)
Value = 5.0	Label = single and never been married
Value = 9.0	Label = not answered

Pos. = 6 Variable = **gor** Variable label = **government office region**

This variable is *numeric*, the SPSS measurement level is **NOMINAL**

Value label information for gor

Value = 1.0	Label = north east
Value = 2.0	Label = north west (incl. the old 'merseyside' region)
Value = 4.0	Label = yorkshire & humber
Value = 5.0	Label = east midlands
Value = 6.0	Label = west midlands
Value = 7.0	Label = south west
Value = 8.0	Label = eastern
Value = 9.0	Label = london
Value = 10.0	Label = south east
Value = 11.0	Label = wales
Value = 12.0	Label = scotland

Pos. = 7 Variable = **gor2** Variable label = **government office region (grouped)**

This variable is *numeric*, the SPSS measurement level is **NOMINAL**

Value label information for gor2

Value = 1.0	Label = england
Value = 2.0	Label = scotland
Value = 3.0	Label = wales

Pos. = 5 Variable = **stdreg** Variable label = **standard region**

This variable is *numeric*, the SPSS measurement level is **NOMINAL**

Value label information for stdreg

Value = 1.0	Label = north
Value = 2.0	Label = yorkshire & humberside
Value = 3.0	Label = east midlands
Value = 4.0	Label = east anglia
Value = 5.0	Label = south east (excl. london)
Value = 6.0	Label = south west

Value = 7.0	Label = west midlands
Value = 8.0	Label = north west
Value = 9.0	Label = wales
Value = 10.0	Label = scotland
Value = 11.0	Label = greater london

Pos. = 949 Variable = london Variable label = !respondent lives in greater london (as defined by g.o.r.)

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for london

Value = 0.0	Label = elsewhere
Value = 1.0	Label = greater london

Pos. = 207 Variable = hetnewp Variable label = new het. sex partner, last year, (if had 1 het. partner in last year)

This variable is *numeric*, the SPSS measurement level is *SCALE*

Value label information for hetnewp

Value = 1.0	Label = yes
Value = 2.0	Label = no
Value = -1.0	Label = not applicable
Value = 9.0	Label = not answered

Pos. = 663 Variable = het Variable label = summary variable: had het. sex. intercourse aged 13+

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for het

Value = 0.0	Label = no het. sex. intercourse ever/since age 13
Value = 1.0	Label = het. sex. intercourse aged 13+
Value = 9.0	Label = unclassifiable

Pos. = 664 Variable = hom Variable label = summary variable: had hom. sex.experience ever

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for hom

Value = 0.0	Label = no hom. sex. experience ever
Value = 1.0	Label = hom. sex. experience ever
Value = 9.0	Label = unclassifiable

Pos. = 944 Variable = hetnew2 Variable label = 1+ new het. sex partners, last year

This variable is *numeric*, the SPSS measurement level is *SCALE*

Value label information for hetnew2

Value = 0.0	Label = 0 new het. sex partners
Value = 1.0	Label = 1+ new het. sex partner(s)
Value = 99.0	Label = not answered
Value = -1.0	Label = not applicable

Pos. = 665 Variable = scelig Variable label = eligible for self-completion questionnaire

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for scelig

Value = 1.0	Label = eligible
Value = 2.0	Label = not eligible

Pos. = 993 Variable = sexcomp Variable label = 'sexual competence' at 1st het. sex. intercourse

This variable is *numeric*, the SPSS measurement level is *SCALE*

Value label information for sexcomp

Value = 0.0	Label = not competent
Value = 1.0	Label = competent
Value = -1.0	Label = not applicable
Value = 9.0	Label = not answered

Ordinal variable

Ordinal means the values are categories **with a meaningful order**, but the distance between categories is not necessarily equal.

Pos. = 494 **Variable = agrp** **Variable label = respondent's age at interview, years, grouped**

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for agrp

Value = 1.0	Label = 16-24
Value = 2.0	Label = 25-34
Value = 3.0	Label = 35-44

Pos. = 495 **Variable = agrp2** **Variable label = respondent's age at interview, years, group**

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for agrp2

Value = 1.0	Label = 16-19
Value = 2.0	Label = 20-24
Value = 3.0	Label = 25-29
Value = 4.0	Label = 30-34
Value = 5.0	Label = 35-39
Value = 6.0	Label = 40-44

Pos. = 666 **Variable = hettot** **Variable label = no. of het. sex partners, life, grouped**

This variable is *numeric*, the SPSS measurement level is *SCALE*

Value label information for hettot

Value = 0.0	Label = 0
Value = 1.0	Label = 1
Value = 2.0	Label = 2
Value = 3.0	Label = 3-4
Value = 4.0	Label = 5-9
Value = 5.0	Label = 10+
Value = 9.0	Label = not answered
Value = -1.0	Label = not applicable

Pos. = 667 **Variable = het5yrg** **Variable label = no. of het. sex partners, last 5 years, grouped**

This variable is *numeric*, the SPSS measurement level is *SCALE*

Value label information for het5yrg

Value = 0.0	Label = 0
Value = 1.0	Label = 1
Value = 2.0	Label = 2
Value = 3.0	Label = 3-4
Value = 4.0	Label = 5-9
Value = 5.0	Label = 10+
Value = 9.0	Label = not answered
Value = -1.0	Label = not applicable

Pos. = 668 **Variable = het1yrg** **Variable label = no. of het. sex partners, last year, grouped**

This variable is *numeric*, the SPSS measurement level is *SCALE*

Value label information for het1yrg

Value = 0.0	Label = 0
Value = 1.0	Label = 1
Value = 2.0	Label = 2
Value = 3.0	Label = 3-4
Value = 4.0	Label = 5+
Value = 9.0	Label = not answered
Value = -1.0	Label = not applicable

Pos. = 669 **Variable = het3mntg** **Variable label = no. of het. sex partners, last 3 months, grouped**

This variable is *numeric*, the SPSS measurement level is *SCALE*

Value label information for het3mntg

Value = 0.0	Label = 0
Value = 1.0	Label = 1
Value = 2.0	Label = 2+
Value = -1.0	Label = not applicable
Value = 9.0	Label = not answered

Pos. = 945 **Variable = hetnew3** **Variable label = no. of new het. sex partners, last year, grouped**

This variable is *numeric*, the SPSS measurement level is *SCALE*

Value label information for hetnew3

Value = 0.0	Label = 0 new het. sex partners
Value = 1.0	Label = 1 new het. sex partner
Value = 2.0	Label = 2+ new het. sex partners
Value = 99.0	Label = not answered
Value = -1.0	Label = not applicable

Pos. = 982 **Variable = educ** **Variable label = left school at 16 with/without passing any exams/gaining any qualifications**

This variable is *numeric*, the SPSS measurement level is *SCALE*

Value label information for educ

Value = 1.0	Label = left school at 16 without passing any exams/gaining quals
Value = 2.0	Label = left school at 16 having passed some exams/gained some quals
Value = 3.0	Label = left school aged 17+
Value = -1.0	Label = not applicable
Value = 9.0	Label = not answered/don't know

Pos. = 983 **Variable = educ2** **Variable label = left school at 16 with/without passing any exams/gaining any qualifications**

This variable is *numeric*, the SPSS measurement level is *SCALE*

Value label information for educ2

Value = 1.0	Label = left school at 16 without passing any exams/gaining quals
Value = 2.0	Label = left school aged 16+ with some exams/gained some quals
Value = -1.0	Label = not applicable
Value = 9.0	Label = not answered/don't know

Pos. = 984 **Variable = educ3** **Variable label = highest academic/non-academic qualification, grouped**

This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for educ3

Value = 1.0	Label = degree
Value = 2.0	Label = higher education, <degree level but a-level/equivalent
Value = 3.0	Label = gcse, o-level or equivalent
Value = 4.0	Label = foreign or other
Value = 5.0	Label = none
Value = 9.0	Label = not answered/don't know

Pos. = 981 Variable = exams2 Variable label = highest academic qualification, grouped
This variable is *numeric*, the SPSS measurement level is *NOMINAL*

Value label information for exams2

Value = 1.0	Label = degree level qualification
Value = 2.0	Label = a-levels/as-levels/slc higher grade
Value = 3.0	Label = o-level/other
Value = 4.0	Label = none
Value = 9.0	Label = not answered/don't know
Value = -1.0	Label = none

Note: Values such as -1, 9, and 99 represent "not applicable" or "not answered" and will be treated as missing values in the analysis.

6. Define the analysis required to answer the RQ and explain why you have chosen this particular analysis.

Weighted negative binomial regression is useful because the outcome is a **count** and data come from a **survey with weights**. The research question is about whether the increase in heterosexual partner counts over time differs by gender, so the key term is:

year × gender

Why not use linear regression?

Linear regression assumes the outcome behaves like a continuous normal variable. But partner counts usually do not. They are often: 0, 1, 2, 3, 4, 5, ...

They are also usually right-skewed (most values are small, but a few values are very large). So linear regression is not ideal.

What about Poisson regression?

Poisson regression is better because it is for count data. But Poisson assumes: *mean ≈ variance*

In sexual partner data, the variance is usually much larger than the mean. That is called **overdispersion**. Poisson is too strict.

```
... Mean: 1.4255991083039197
Variance: 6.023213495495001
Variance / Mean: 4.225040167611362
```

The variance is more than **4 times larger** than the mean. That means the outcome is **overdispersed**. This strongly supports **negative binomial regression**.

Why negative binomial regression

Negative binomial regression is better because it allows: *variance > mean*

With weights, our results better represent the target population, that's why we use *total_wt*.

Dependent variable: *het1yr* not *het1yrg* because the grouped variable does not specify the exact number of partners.

7. List all assumptions and how you addressed them.

1. The most important assumption is that **the dependent variable is an appropriate count outcome for negative binomial regression**. This was addressed by using *het1yr_clean*, the raw number of heterosexual partners in the last year, instead of the grouped variable *het1yrg*. Invalid codes such as "not applicable" and "not answered" were recoded as missing so that only valid non-negative counts were included.
2. A second important assumption is **overdispersion**. Negative binomial regression is appropriate when the variance of the count outcome is larger than the mean. This was addressed by choosing negative binomial regression instead of Poisson regression, because partner-count data are typically right-skewed and overdispersed.
3. The model also assumes that **relevant confounders are controlled for**. This was addressed by including age, education, marital status, and region as control variables.
4. Finally, because the data come from a survey, **weights should be applied**. This was addressed by using the survey weight in the regression, although full survey design features such as PSU and strata were not fully modeled.

8. List the null and alternative hypothesis.

The **null hypothesis** is that gender does not significantly affect the change in the number of heterosexual partners over time. In other words, the change in reported heterosexual partner counts from 2000 to 2010 is the same for men and women.

$$H_0: \beta_{\text{gender} \times \text{year}} = 0$$

The **alternative hypothesis** is that gender does significantly affect the change in the number of heterosexual partners over time. This means that the change in reported heterosexual partner counts from 2000 to 2010 differs between men and women.

$$H_1: \beta_{\text{gender} \times \text{year}} \neq 0$$

In this analysis, the hypothesis is tested using the interaction term between survey year and gender in the weighted negative binomial regression model.

9. Provide the results of your analysis and answer the research question.

The results most relevant to the research question are the effects of survey year, gender, and especially the interaction between survey year and gender.

survey year effect: `C(year2010) [T.1] coef = -0.0256 P>|z| = 0.495`

The main effect of survey year was not statistically significant, $b = -0.026$, $p = .495$. This means that, for the reference group, men, there was no significant change in the expected number of heterosexual partners from 2000 to 2010.

gender effect: `C(female) [T.1] coef = -0.2227 P>|z| = 0.000`

The main effect of gender was statistically significant, $b = -0.223$, $p < .001$, IRR = 0.800. This shows that women reported about 20% lower expected heterosexual partner counts than men in the reference year.

gender × year interaction: `C(year2010) [T.1]:C(female) [T.1] coef = 0.1100 P>|z| = 0.004`

Most importantly, the gender × year interaction was statistically significant, $b = 0.110$, $p = .004$, IRR = 1.116. This indicates that the change in heterosexual partner counts from 2000 to 2010 differed significantly by gender. Specifically, the change over time was about 11.6% more positive for women than for men.

Therefore, the research question can be answered as follows: gender was a significant factor in the change in reported heterosexual partner counts over time. Men did not show a significant change between the two survey years, while women showed a more positive change.

10. Try to explain the impact of the result within the context of the data.

The result suggests that gender played an important role in how reported heterosexual partner counts changed between the two survey waves. The significant gender × year interaction indicates that the change from 2000 to 2010 was not the same for men and women. Specifically, women showed a more positive change over time compared with men.

Within the context of the data, this means that although women reported lower expected partner counts than men in the reference year, the gap between men and women appears to have become smaller over time. Men did not show a statistically significant change between the two survey years, while women's reported partner counts increased more strongly.

This result may reflect changes in sexual behaviour, social norms, or reporting patterns over time. For example, women may have become more likely to report heterosexual partners openly in the later survey wave, or actual behaviour may have changed. However, because the data are cross-sectional survey data, the result should be interpreted as an association rather than proof of a causal effect.

Overall, the impact of the result is that gender should be considered when studying changes in sexual behaviour over time. Looking only at the overall change between survey years would miss the fact that the pattern differs between men and women.